

DATA 119 Group Project Update

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Data Cleaning:

So far, we have only been looking at individual Clash Royale cards, not decks yet.

To collect the data, we combined four different datasets:

- <https://developer.clashroyale.com>: the official API, containing all the latest cards.
- <https://www.kaggle.com/datasets/rodsaldanha/clash-royale-matches>: contains the damage and hitpoints of each card for different levels. Since all cards are fixed at level 11 in competitive play, we will only use those stats.
- <https://www.kaggle.com/datasets/niteshkakkar/clash-royal-cards-data>: contains hitpoints, card type, mobility, target, attack type, group, and usage rate.
- <https://www.kaggle.com/datasets/hrish4/clash-royale-cards-data>: contains the win rate of all the cards.

We first merged all four dataframes together and saved it as a csv. We then loaded the csv into another notebook and examined its missing values. The maxEvolutionLevel column contained 1, 2, 3, or null. Upon looking at a few rows, it was clear that this is actually a categorical variable; 1 represents an evolution card, 2 represents a hero card, 3 represents a card that is both, and null means the card is neither (in Clash Royale, an evolution card cycles through its regular and buffed state during a match, whereas a hero card can be buffed everytime using elixir). So all 73 missing values were actually just regular cards. Thus, we replaced the numerical values and nulls with the appropriate strings and renamed the variable as “wild” for wildcard.

The damage₁₁ variable had 29 missing values; some of them are due to the Kaggle dataset being outdated, and some of them are due to unique cards that can have different damages based on interpretation. These were all easily fixed. The hitpoints variable had 20 missing values; only one (the Spirit Empress) was due to it being a new card. The other 19 missing values were simply because the cards are spells such as fireball and arrows, which have no hitpoints associated with them. The variable was also renamed as “hitpoints₁₁” for clarity.

All the other columns only had one or two missing values. The ones that were due to Spirit Empress were easily fixed. The rest were due to a card called “mirror.” The mirror card deploys the troop you last deployed, so its stats are hard to define since it is dependent on the previous card played. For future analysis, the mirror card will be omitted as its mechanics are different from the rest of the cards, and it also has one of the lowest usage and win rates.

EDA:

To begin visualizing the data, we graphed a few univariate and bivariate distributions.

The scatterplot of damage vs. health indicates that few cards are high in both stats, and high-health cards may be more common than high-damage cards.

The most common card type is troop, which is more than half of all cards; the rest is a mix of building and spell cards.

While card win rate ranges from around 43% to 61%, the card with the highest usage rate (20% and higher) fall closer to the middle of the win rate range than the high end, suggesting that the most common cards may not necessarily be the most successful.